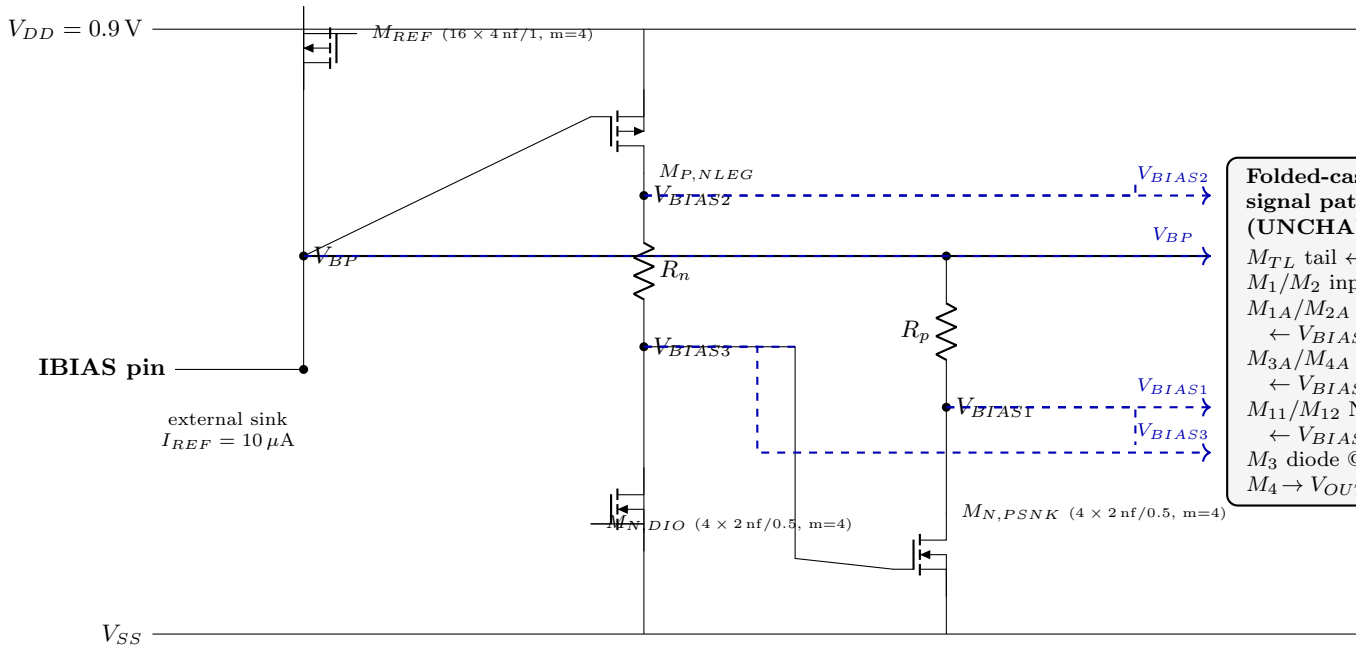


R_n-degenerated wide-swing cascode bias - sky130A LVT, $V_{DD} = 0.9\text{ V}$

Col 1: master

Col 2: NMOS leg

Col 3: PMOS leg



**Folded-cascode
signal path
(UNCHANGED)**

M_{TL} tail $\leftarrow V_{BP}$
 M_1/M_2 input pair
 M_{1A}/M_{2A} NMOS casc.
 $\leftarrow V_{BIAS2}$
 M_{3A}/M_{4A} PMOS casc.
 $\leftarrow V_{BIAS1}$
 M_{11}/M_{12} NMOS bot.
 $\leftarrow V_{BIAS3}$
 M_3 diode @ f_L
 $M_4 \rightarrow V_{OUTP}$